

Part A - Understanding One-Person Company Systems

1. The Company Concept Of One Person:

The entrepreneurially solo venture is distinguished by one person, known as the Founder, who acts as both the owner, manager, and operator of the business (Cieslik & Van Stel, 2022). Unlike conventional businesses, which require more than one department and workers, the profit-making venture uses digital tools in organizing the business (Rezazadeh & Bonehgazy, 2025). The firm uses automated computer software and artificial intelligence agents to enhance the efficiency of humans in executing their regular tasks in the business operations. Digital platforms provide the company with the ability to manage different aspects such as finances, marketing, and service functions through one comprehensive interaction process, thus freeing the Founder from monotonous work and enabling him to focus on decision-making while routine work is performed automatically (Rezazadeh & Bonehgazy, 2025).

2. One-Person Company Architecture:

Under the single operator architecture, the Founder is viewed as the final authority when decisions are made (Rezazadeh & Bonehgazy, 2025). Under this particular system, the Founder will be interacting with several digital agents where each one focuses on some different area in the company (Sabarish, 2025). The digital agents are basically software modules designed to conduct different tasks, like the housekeeper in the home environment or the cash registrar in business that handles money transactions in accounts and sales. Various processes will be done using these agents where different tasks are assigned to these agents by the Founder. There are various other components under this system like the Founder and agents, the database that holds data, the task manager, and communication component that allows interaction between the Founder and agents (MIT Sloan, 2025; Sabarish, 2025).

3. Multi-Agent Systems Roles:

A multi-agent system involves a number of intelligent agents that work together as a team to achieve a shared goal (K., 2026). In this regard, each individual agent works like an independent entity, but they interact for effective coordination of their activities to meet organizational objectives (K., 2026). For example, the Finance Agent could be responsible for budgeting and expenditure, the Marketing Agent for the marketing activities, and the Customer Service Agent for attending to customer complaints. Although the various agents are separate entities, interaction among them takes place whenever necessary. Through such cooperation, the whole process becomes systematic and efficient. Through the use of several agents for different operations within the business, simultaneous operations become possible, thus increasing efficiency (MIT Sloan, 2025).

4. The Perspective Of Programming:

From the perspective of programming, the one-man company model may be thought of as a form of object-oriented programming (Leite, 2006). In object-oriented programming, the development of the system is done through classes and objects representing tangible things in the real world (Sabarish, 2025). For example, the founder might be regarded as a class that is bestowed with attributes such as name and the ability to make decisions, whereas the agents could be treated as other classes assigned corresponding duties. The tasks are objects themselves and contain more information about the task itself. Through the use of objects, interfaces, inheritance, encapsulation, and polymorphism, elegant and reusable code can be written. An agent or the set of agents may be inherited from an "Agent" parent class, with each agent showing its own unique behavior (Leite, 2006; Rezazadeh & Bonehgazy, 2025).

Part B – Business Idea and Project Proposal

1. Name of the Business:

This business has been named as Smart Study Tutoring Services.

2. Background of the Business:

Smart Study Tutoring Service is a type of online tutoring company where the emphasis is on quality teaching to instill good studying skills in the students. This business offers tutoring services for science, math, and programming subjects on a one-to-one basis. The services are now available to those students who opt for distance education over the internet. Appointments may be made, and records are kept digitally. The vision statement is to provide quality tutoring services in an affordable manner to the students, which could be managed by an individual effectively.

3. Target Customers:

The target market for this product consists of students from various schools and universities who may be in need of help in their studies. It would include people who are preparing for their exams, those who find it difficult in one specific area, and those who wish to enhance their academic performance. There is also a target market for parents who seek additional education for their kids. Since this is an online platform, it has the capability to reach out to customers across the globe.

4. Why One Person Can Manage This Business:

The venture operates in digital ecosystems and makes use of automated agents, which make it possible for one person to control everything. Activities like scheduling classes, sending out reminders, collecting payments, and handling customer queries can be done automatically. The Founder spends most of their time teaching, making decisions, and monitoring the system. With the help of virtual agents, activities that are repetitive and require much time can be done without human involvement. It is therefore possible to conduct the entire business operation at home through one person.

5. Proposed Agents:

The system will have several agents designed to facilitate its functioning:

- Finance Agent: Deals with the collection of fees, payment disbursements, and receipt statement generation.
- Marketing Agent: Takes care of marketing teachers' requests on social media platforms and other websites.
- Operations Agent: Manages the scheduling and task allocation of tutors.
- Customer Support Agent: Handles students' questions, reminds them, and offers necessary help.

Each agent acts as an expert in a particular domain of the company's operation.

6. Example Interaction:

The Founder interacts with the system through sending the requests and questions to the designated agents. For example, the Founder may ask the Operations Agent to book tutoring for a certain student. The Marketing Agent might come up with ideas on how to attract more students in light of existing trends. The Finance Agent can provide the Founder with the monthly earnings report, whereas the Customer Support Agent might inform the Founder about questions raised by students. Depending on those agents, it falls onto the Founder's shoulders to make the ultimate decision regarding pricing, service, and session changes.

Part C – System Planning and Design

1. System Architecture:

The system is built to be a Multi-Agent System, where the Founder acts as the main decision-maker. The system follows a modular design pattern, wherein the Founder designs the task, whereas different Digital Agents perform the task. These agents hold domain knowledge and generate Recommendations; however, the final decision regarding the business rests with the Founder.

2. Multi-Agent Structure:

As per the Smart Study Tutoring Services, the agents shall be:

- The Finance Agent: He will collect fees, pay out fees, and keep track of all receipts.
- The Marketing Agent: She will deal with advertisement inquiries and manage social media sites to recruit more students.
- The Operations Agent: He will control scheduling and conducting of the sessions.
- The Customer Support Agent: She will answer student queries and remind about session timing.

3. System Classes:

The above-described system architecture uses the following essential classes for implementation:

-Founder class:

It describes the actor who is able to create tasks and receive reports.

-Agent class/es:

They describe the behavior of the digital assistants and the requirements for such behavior.

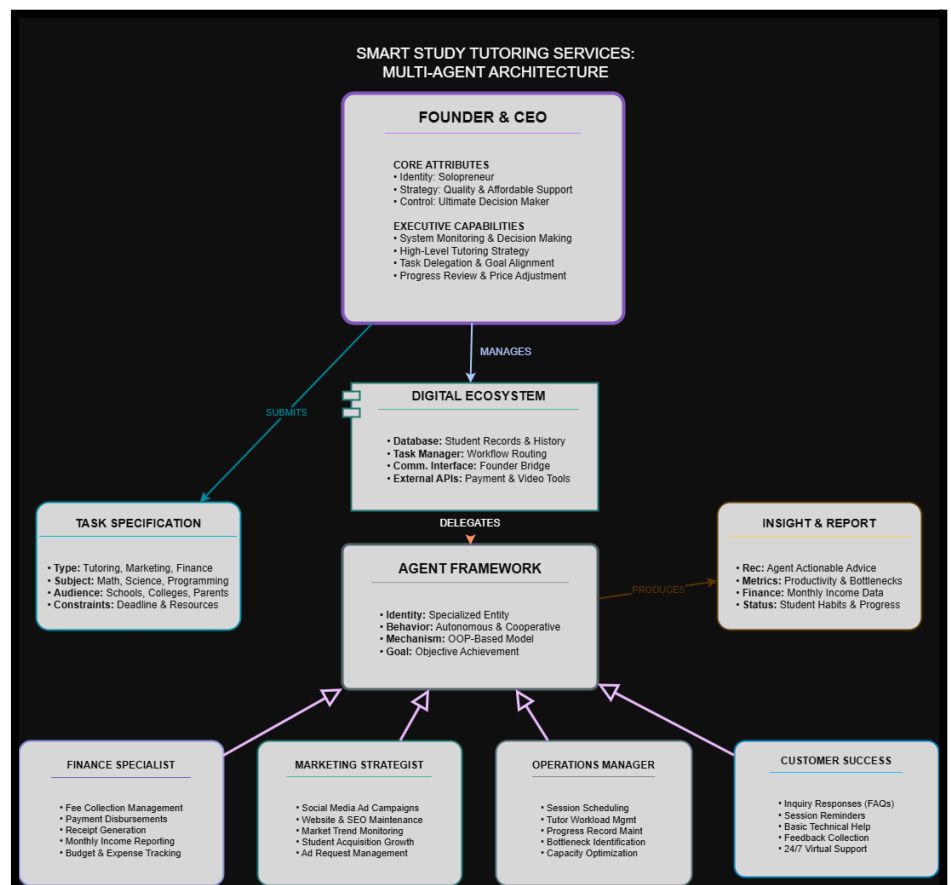
-Task object:

It represents actions/queries made by the founder within a certain business.

-Recommendation/Report:

It is an output produced by the agent as a result of the processing of data received in a task.

4. The Diagram Of Class UML:



Part D – System Implementation

GitHub Repository Link:

[\[https://github.com/KyDo0/INTE130-OnePersonCompany.git\]](https://github.com/KyDo0/INTE130-OnePersonCompany.git)

Python Code Link:

[\[https://github.com/KyDo0/INTE130-OnePersonCompany/blob/main/implementation.py\]](https://github.com/KyDo0/INTE130-OnePersonCompany/blob/main/implementation.py)

Google Colab Link: [\[https://colab.research.google.com/drive/15jUKbp-5h__RKeobH0O4Cmtaw_D40tcE?usp=sharing\]](https://colab.research.google.com/drive/15jUKbp-5h__RKeobH0O4Cmtaw_D40tcE?usp=sharing)

Certificate Link:

[\[https://github.com/KyDo0/INTE130-OnePersonCompany/blob/main/certificate.pdf\]](https://github.com/KyDo0/INTE130-OnePersonCompany/blob/main/certificate.pdf)